# The Periodic Table as $\xi$-Geometry: Radial Mass Ladder and Angular Shell Structure from a Common Origin

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## Contents

## Chapter A

# The Periodic Table as $\xi$-Geometry

## Abstract

The periodic table of the elements and the particle-mass ladder of T0 theory share the same geometric quantum number $n$, but in different roles. Particle masses follow a *radial* exponential scaling $m_n \sim \xi^{-n} \cdot m_P$, while the shell structure of the periodic table obeys an *angular* quadratic degeneracy $g_n = 2n^2$. Both arise from the three-dimensional lattice geometry of T0 theory: $\xi$ is the packing deficit of the 3D lattice (radial direction), $2n^2$ is the degeneracy of the $n$-th spherical shell in the same lattice (angular direction). The Rydberg energy lies exactly on rung 7 of the $\xi$-power ladder: $R_\infty = \frac{m_e \alpha^2}{2} \approx \xi^{6.956} \cdot E_P$. The duality principle $E_n \cdot g_n = 2R_\infty =$ const connects the $1/n^2$ energy ladder with the $n^2$ filling ladder and is a geometric equipartition. The periodic table is the angular cross-section of the $\xi$-geometry at rung $\xi^{\sim 7}$ — analogously, the lepton masses are the radial cross-section at rungs $\xi^{4.9}$–$\xi^{5.8}$.

## A.1 Introduction and Problem Statement

T0 theory (Time-Mass Duality / FFGFT) derives all physical constants from the geometric parameter

$$\xi = \frac{4}{30\,000} = 1.\overline{3} \times 10^{-4} \tag{A.1}$$

[**?**, **?**, **?**]. The particle masses of the Standard Model organise themselves on an exponential ladder

$$m_n \;\sim\; \xi^{-n} \cdot m_P, \qquad n = 1, 2, 3, \ldots \tag{A.2}$$

where $n$ is the T0 quantum number and $m_P$ is the Planck mass.

The periodic table of the elements, on the other hand, shows a very different pattern: shell capacities follow

$$g_n \;=\; 2n^2 \;=\; 2,\; 8,\; 18,\; 32,\; 50,\; \ldots \tag{A.3}$$

and binding energies scale as $1/n^2$. At first glance, particle masses and the periodic table appear unrelated.

**Gemini's observation** (2026): the periodic table lies "one level higher" than the particle masses — both have quantum numbers, but what common geometric origin connects their patterns?

The answer of this document: *T0 geometry has two axes — a radial one ($\xi^n$ scaling) and an angular one ($2n^2$ degeneracy), both arising from the three-dimensional lattice structure with packing deficit $\xi$.*

## A.2 The $\xi$-Power Ladder and Its Rungs

Previous documents in the T0 series have built up the $\xi$-power ladder step by step. Table **??** shows the current state including the new atomic rung.

**Table A.1:** Complete $\xi$-power ladder (as of March 2026)

| Rung | $\xi$ power | Physical content | Document |
|---|---|---|---|
| $0$ | $\xi^0 = 1$ | Planck scale, $L_0 = \xi \cdot \ell_P$ | Doc. 009 |
| $1/2$ | $\xi^{1/2} \approx 10^{-2}$ | $\alpha_W$, $m_e/m_\mu$, PPLN $\Delta n$ | Doc. 041, 167 |
| $1$ | $\xi^1 \approx 10^{-4}$ | $\alpha_{\text{EM}} \sim \xi \cdot E_0^2$ | Doc. 011 |
| $4$ | $\xi^4 \approx 10^{-16}$ | CMB temperature $T_{\text{CMB}}$ | Doc. 166 |
| $23/4$ | $\xi^{5.775}$ | Electron mass $m_e/m_P$ | Doc. 056 |
| $\sim$ **7** | $\xi 0^{6.956}$ | **Rydberg energy $R_\infty$, periodic table** | **this doc.** |
| $10$ | $\xi^{10} \approx 10^{-43}$ | Hubble constant $H_0$ | Doc. 165 |

The Rydberg energy lies between rungs 6 and 7:

$$\frac{R_\infty}{E_P} = \frac{m_e}{m_P} \cdot \frac{\alpha^2}{2} = \xi^{5.775} \cdot \xi^{1.181} = \xi^{6.956} \approx \xi^7 \tag{A.4}$$

The exponent is the sum of two independent T0 quantities: the electron mass (rung $5.775$, Doc. 056) and the fine-structure constant (rung $1.181$, Doc. 011).

## A.3 Two Axes of $\xi$-Geometry

### Radial Axis: Exponential Mass Ladder

Particle masses organise themselves along the *radial* axis of the T0 lattice. The fundamental relation is [**?**]:

$$m_n \sim \xi^{-n} \cdot m_P, \qquad n = 1, 2, 3, \ldots \tag{A.5}$$

Each step on this ladder multiplies the mass by $\xi^{-1} = 7500$. The leptons occupy the rungs (Doc. 056, 122):

$$m_e : \quad \xi^{5.775} \cdot m_P \tag{A.6}$$
$$m_\mu : \quad \xi^{5.177} \cdot m_P \tag{A.7}$$
$$m_\tau : \quad \xi^{4.861} \cdot m_P \tag{A.8}$$

The mass ratio $m_e/m_\mu \propto \xi^{1/2}$ (Doc. 041) lies exactly on the electroweak exponent — the half-rung of the $\xi$-ladder.

### Angular Axis: Quadratic Degeneracy Ladder

The *angular* axis of the same lattice determines the degeneracy of spherical shells. In 3D space, the $n$-th principal shell has:

$$g_n = 2\sum_{l=0}^{n-1}(2l+1) = 2n^2 \tag{A.9}$$

The factor 2 comes from spin $m_s = \pm\frac{1}{2}$; the sum is pure angular-momentum algebra in 3D space. Table **??** shows the first four shells.

**Table A.2:** Shell structure and the periodic table

| $n$ | $g_n = 2n^2$ | Period length | Noble gas $Z$ | Period number |
|---|---|---|---|---|
| 1 | 2 | 2 | He (2) | 1 |
| 2 | 8 | 8 | Ne (10) | 2, 3 |
| 3 | 18 | 18 | Ar (18) | 4, 5 |
| 4 | 32 | 32 | Kr (36) | 6, 7 |

### The Connection through Quantum Number $n$

The same quantum number $n$ appears in both systems, but with different functional roles:

$$\text{Radial (masses):} \quad m_n \sim \xi^{-n} \cdot m_P \quad \text{(exponential)} \tag{A.10}$$
$$\text{Angular (shells):} \quad g_n = 2n^2 \quad \text{(quadratic)} \tag{A.11}$$

This is not coincidental. In T0 lattice geometry, $n$ corresponds to the *discrete radius in the lattice*:

- Radially: $n$ steps outward from the lattice $\Rightarrow$ mass scales as $\xi^{-n}$ (geometric sequence)
- Angularly: the lattice points on the sphere of radius $n$ carry degeneracy $2n^2$ (combinatorial sequence)

## A.4 The Duality Principle

### Energy Ladder and Filling Ladder Are Dual

On the atomic scale (hydrogen-like atoms):

$$\text{Energy ladder:} \quad E_n = -\frac{R_\infty}{n^2} \quad \text{(falls as } 1/n^2\text{)} \tag{A.12}$$

$$\text{Filling ladder:} \quad g_n = 2n^2 \quad \text{(grows as } n^2\text{)} \tag{A.13}$$

Their product is exactly constant:

$$\boxed{E_n \cdot g_n = \frac{R_\infty}{n^2} \cdot 2n^2 = 2R_\infty = \text{const}} \tag{A.14}$$

This is geometric *equipartition*: every shell contributes the same total energy $2R_\infty$ to the Rydberg scale, regardless of $n$. The $1/n^2$ energy ladder and the $n^2$ filling ladder are *inversely dual*.

### Comparison with the Mass Ladder

Table **??** summarises all three ladders.

**Table A.3:** Three ladders — one geometric origin

| System | Ladder | Function of $n$ | Type |
|---|---|---|---|
| Particle masses | $m_n \sim \xi^{-n} \cdot m_P$ | exponential | radial |
| Shell energy | $E_n = R_\infty/n^2$ | harmonic | angular |
| Shell filling | $g_n = 2n^2$ | quadratic | angular |
| Duality: $E_n \cdot g_n = 2R_\infty = \text{const}$ | | | |

## A.5 The Role of Fractal Dimension

In T0 (Doc. 009, 133):

$$D_f = 3 - \xi = 2.999867 \tag{A.15}$$

Space dimension is almost, but not exactly, 3.

In the limit $D_f \to 3$ ($\xi \to 0$):

- $g_n = 2n^2$ is *exact* (pure spherical symmetry in the $\mathbb{Z}^3$ lattice)
- the shell structure of the periodic table is exact

For $D_f = 3 - \xi < 3$ there is a minimal correction:

$$\delta g_n \sim \xi \cdot n^2 \tag{A.16}$$

For $n = 2$: $\delta g_2 \approx \xi \cdot 4 \approx 5 \times 10^{-4}$ (entirely negligible).

The periodic table is therefore exact because $D_f \approx 3$ to four decimal places. The tiny deviation $\xi \approx 1.3 \times 10^{-4}$ makes space "almost-integer-dimensional" — and precisely this deviation generates all the physics of particle masses and fundamental constants.

## A.6 Numerical Verification

### Rydberg Energy on the $\xi$-Ladder

$$R_\infty = \frac{m_e \alpha^2}{2} = 13.5984 \text{ eV} \tag{A.17}$$

$$\frac{R_\infty}{E_P} = \frac{m_e}{m_P} \cdot \frac{\alpha^2}{2} = \xi^{5.775} \cdot \xi^{1.181} = \xi^{6.956} \tag{A.18}$$

**Table A.4:** Rydberg energy: T0 derivation vs. measurement

| Quantity | Value | $\xi$-exponent |
|---|---|---|
| $m_e/m_P$ | $4.185 \times 10^{-23}$ | 5.775 |
| $\alpha^2/2$ | $2.663 \times 10^{-5}$ | 1.181 |
| $R_\infty/E_P$ | $1.114 \times 10^{-27}$ | 6.956 |
| $R_\infty$ (T0, approx.) | $\approx 9.1$ eV ($\xi^7$) | 7.000 |
| $R_\infty$ (measured) | 13.598 eV | 6.956 |

### Duality Relation: Numerical Verification

$$E_1 \cdot g_1 = 13.60 \text{ eV} \cdot 2 = 27.20 \text{ eV} = 2R_\infty \tag{A.19}$$

$$E_2 \cdot g_2 = 3.40 \text{ eV} \cdot 8 = 27.20 \text{ eV} = 2R_\infty \tag{A.20}$$

$$E_3 \cdot g_3 = 1.51 \text{ eV} \cdot 18 = 27.18 \text{ eV} \approx 2R_\infty \tag{A.21}$$

$$E_4 \cdot g_4 = 0.85 \text{ eV} \cdot 32 = 27.20 \text{ eV} = 2R_\infty \tag{A.22}$$

The duality relation (**??**) holds exactly for all $n$.

## A.7 The Periodic Table in the T0 Hierarchy

| $\xi$-rung | Physics | Axis |
|---|---|---|
| $\xi^0 = 1$ | Planck scale ($L_0 = \xi \cdot \ell_P$) | — |
| $\xi^{1/2}$ | Electroweak: $\alpha_W$, $m_e/m_\mu$ | radial |
| $\xi^1$ | $\alpha_{\text{EM}}$, EM coupling | radial |
| $\xi^4$ | CMB temperature $T_{\text{CMB}}$ | radial |
| $\xi^{5.8}$ | Particle masses (leptons, quarks) | radial |
| $\xi^{6.956}$ | **Rydberg $R_\infty$, periodic table** | **both** |
| $\xi^{10}$ | Hubble constant $H_0$ | radial |

**Figure A.1:** Position of the periodic table in the complete $\xi$-hierarchy

The special feature of the atomic rung $\xi^{\sim 7}$: here *both* axes act. The radial axis determines the energy scale $R_\infty$; the angular axis determines the pattern $2n^2$. The periodic table is the visible result of both axes acting together.

## A.8 Predictions and Open Questions

### Angular Corrections from $D_f < 3$

The fractal dimension $D_f = 3 - \xi$ leads to minimal corrections of the exact $2n^2$ rule:

$$g_n^{(\mathrm{T0})} = 2n^2 \cdot (1 - \xi \cdot f(n)) \tag{A.23}$$

where $f(n)$ is a dimensionless function of $n$ (still to be derived). For $n \leq 4$ the correction is $< 10^{-3}$ and not measurable.

### Molecular Binding Energies

Chemical binding energies lie in the range $1$–$10$ eV, hence also on rung $\xi^7$. It remains to be checked whether specific binding energies lie on rational multiples of $\xi^{7/2} \cdot R_\infty$.

### Heavy Elements and Relativistic Corrections

For $Z \gg 1$ relativistic effects become important. In T0 these are connected with the $\xi^{1/2}$ electroweak rung. A systematic analysis of ionisation energies of heavy elements ($Z > 54$) might reveal $\xi$-matches at the half-rung $\xi^{7-1/2} = \xi^{13/2}$.

## A.9 Summary

1. **Energy scale:** $R_\infty/E_P = \xi^{6.956} \approx \xi^7$. The Rydberg energy — and hence all of atomic physics — lies on rung 7 of the $\xi$-power ladder.
2. **Radial axis:** Particle masses follow $m_n \sim \xi^{-n} \cdot m_P$. The exponent $n$ is the discrete lattice radius in the radial $\xi$-direction.
3. **Angular axis:** Shell capacities follow $g_n = 2n^2$. The index $n$ is the same discrete lattice radius, now acting as a shell index in the angular 3D-sphere direction.
4. **Duality principle:** $E_n \cdot g_n = 2R_\infty =$ const. The $1/n^2$ energy ladder and the $n^2$ filling ladder are inversely dual. Every shell contributes the same total energy $2R_\infty$ to the $\xi^7$ scale.
5. **Fractal dimension:** $D_f = 3 - \xi \approx 3$ makes $2n^2$ exact (to $\sim 10^{-4}$). The periodic table is exact because space is almost-integer-dimensional.
6. **Common origin:** Particle masses and the periodic table arise from the same 3D lattice geometry with packing deficit $\xi$. The periodic table is the *angular cross-section* of $\xi$-geometry at rung $\xi^{\sim 7}$; the lepton masses are the *radial cross-section* at rungs $\xi^{4.9}$–$\xi^{5.8}$.

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